Development of a Synthetic Epidemiological Benchmark Dataset for Evaluating Causal Inference Methods

Technical Report

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Contents

1. Abstract

Synthetic datasets have become an essential component of modern causal inference research because they allow researchers to evaluate statistical methods under controlled environments where the true causal effect is known. Unlike observational datasets, synthetic benchmark datasets provide complete access to potential outcomes, treatment assignment mechanisms, and underlying data generating processes.

This report presents the methodology adopted for constructing a synthetic epidemiological benchmark dataset based on an extended Susceptible–Exposed–Infectious–Recovered (SEIR) compartmental model. The simulator incorporates realistic demographic characteristics, behavioural factors, treatment assignment through observational confounding, compliance behaviour, and stochastic disease transmission.

The generated dataset consists of 100,000 simulated epidemic scenarios with 24 variables describing demographic characteristics, intervention information, epidemiological measures, potential outcomes, and observed outcomes. The resulting benchmark dataset is specifically designed for evaluating causal inference techniques including Propensity Score Matching (PSM), Inverse Probability Weighting (IPW), Augmented Inverse Probability Weighting (AIPW), Doubly Robust estimators, DoWhy estimators, and machine learning based methods such as T-Learners.

2. Introduction

Randomized controlled trials represent the gold standard for estimating causal effects. However, conducting randomized experiments is frequently expensive, ethically infeasible, or practically impossible in public health applications. Consequently, observational data are commonly analysed using causal inference techniques to estimate intervention effects.

One of the major challenges in evaluating causal inference methods is the absence of ground-truth treatment effects in real observational datasets. Since the true counterfactual outcomes are never observed, the actual performance of an estimator cannot be measured directly.

Synthetic benchmark datasets overcome this limitation by generating observations through a known data generating process (DGP). Because the simulator explicitly constructs both potential outcomes under treatment and no treatment, the true Individual Treatment Effect (ITE), Average Treatment Effect (ATE), and Average Treatment Effect on the Treated (ATT) become available for quantitative evaluation.

The primary objective of this work is therefore not to simulate a real epidemic perfectly, but rather to construct a statistically meaningful benchmark environment for comparing multiple causal inference methods under realistic observational settings.

3. Project Development Roadmap

Project Evolution

This project did not begin directly as a benchmark dataset. It evolved through several iterations after identifying limitations in earlier synthetic data generation approaches. The development process is summarized below.

1. **Literature Review**

   The project began with studying modern causal inference methods including Propensity Score Matching (PSM), Inverse Probability Weighting (IPW), Augmented IPW (AIPW), Doubly Robust Estimation, EconML, Meta-Learners and recent applications in epidemiology.

2. **Initial Synthetic Dataset**

   A basic SEIR epidemic simulator was first constructed. Although useful for testing algorithms, the generated data contained overly simplistic treatment assignment and lacked realistic behavioural variation.

3. **Identification of Limitations**

   After evaluating the initial benchmark, several weaknesses were identified:

   - Treatment assignment was overly deterministic.
   - No behavioural compliance mechanism existed.
   - Limited heterogeneity in intervention effects.
   - Epidemic dynamics were comparatively simple.

4. **Dataset Redesign**

   The data-generating mechanism was redesigned by introducing

   - Behavioural compliance,
   - Contact intensity,
   - Propensity-score based treatment assignment,
   - Time-varying intervention effects,
   - Hidden behavioural heterogeneity,
   - Counterfactual outcomes,
   - Individual treatment effects.

5. **Pilot Validation**

   A small pilot dataset consisting of 100 simulated epidemics was generated. Distributional properties, treatment proportions, correlations and intervention effects were examined before scaling.

6. **Large Scale Benchmark Generation**

   Following satisfactory pilot validation, the final benchmark containing 100,000 simulated epidemic scenarios was generated.

7. **Dataset Validation**

   The generated benchmark was validated using

   - Descriptive statistics
   - Distributional analysis
   - Correlation analysis

- Missing value checks
- Duplicate detection
- Treatment balance
- Counterfactual consistency

8. **Benchmark Comparison**

   Finally, the proposed benchmark was compared against another independently generated synthetic SEIR dataset to evaluate epidemiological variability and causal modelling capability.

9. **Future Direction**

   The final benchmark will be used for evaluating classical causal inference methods together with modern Causal Machine Learning algorithms. Subsequently, these methods will also be validated on real-world observational datasets.

4. Overall Project Workflow

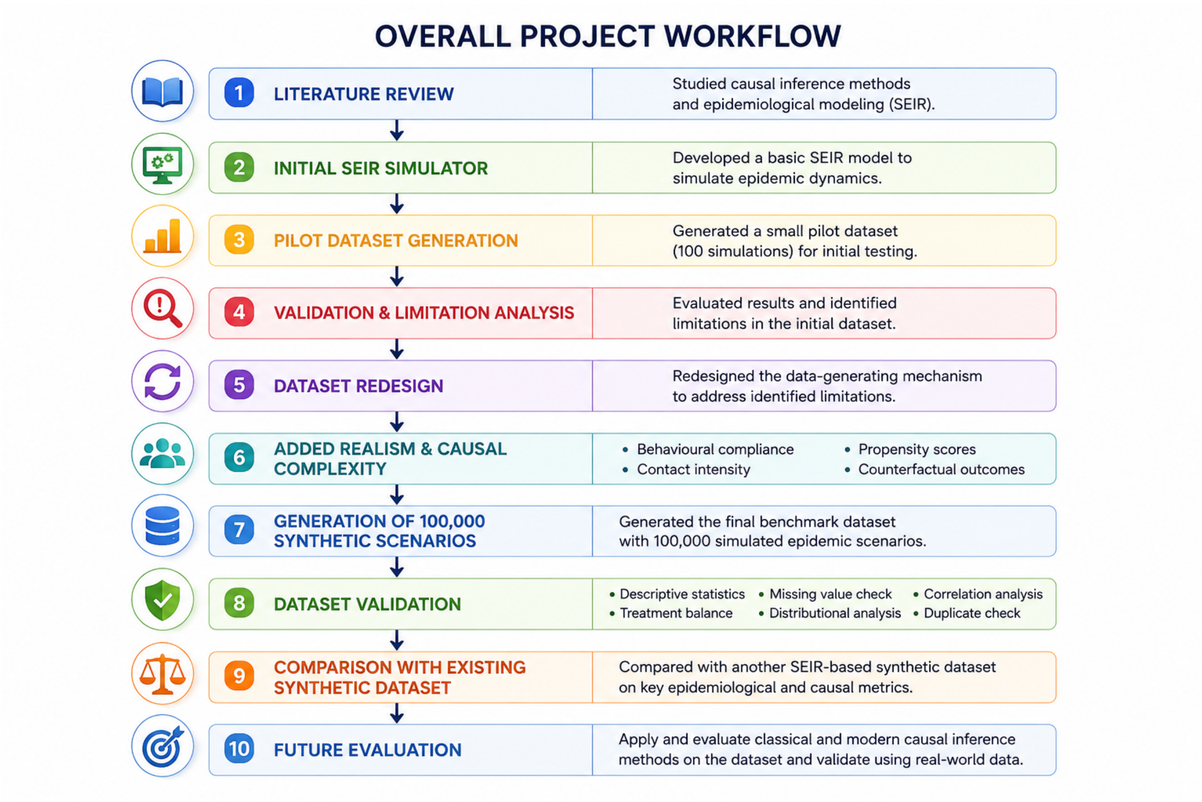


Figure 1: Overall Project Workflow

5. Motivation

The motivation behind constructing a synthetic benchmark dataset arises from three major limitations of publicly available epidemiological datasets.

1. Ground truth causal effects are unknown.
2. Treatment assignment is usually partially observed or unavailable.
3. Counterfactual outcomes cannot be measured.

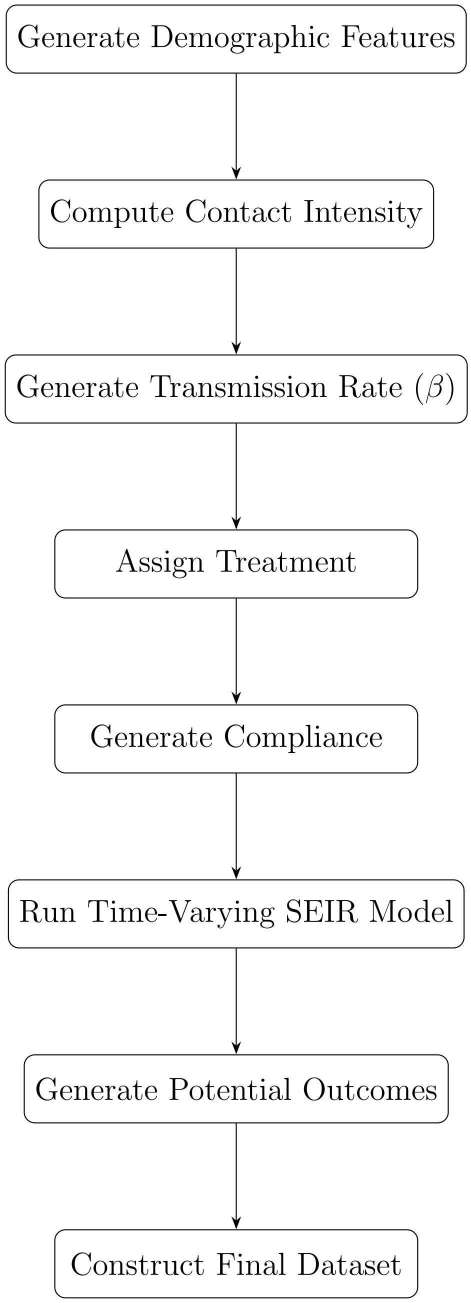
Consequently, comparing causal estimators solely on real-world data becomes difficult because estimation accuracy cannot be quantified.

To address these limitations, the proposed simulator generates complete observational data while retaining full knowledge of the underlying treatment mechanism and potential outcomes. This allows systematic evaluation of different causal inference algorithms using known causal quantities.

Unlike simple toy simulations, the proposed benchmark introduces realistic complexity through demographic heterogeneity, behavioural compliance, nonlinear transmission dynamics, stochastic epidemic evolution, and observational treatment assignment.

6. Overall Simulation Framework

The complete data generation process follows a sequential pipeline in which demographic characteristics influence behavioural interactions, behavioural interactions determine disease transmission, transmission governs epidemic evolution, and treatment modifies disease spread through behavioural compliance.



Each stage contributes to the realism of the benchmark by incorporating observational confounding and heterogeneous treatment effects.

7. Generation of Covariates

The first stage of the simulation generates demographic and epidemiological characteristics representing a hypothetical geographical region.

The generated variables include

- Population Density
- Vaccination Rate

- Human Mobility
- Mask Usage
- Hospital Capacity
- Mean Population Age

These variables are generated independently from continuous probability distributions chosen to represent plausible ranges observed across different districts or administrative regions.

Rather than serving only as descriptive variables, these covariates influence several downstream components of the simulator. Specifically,

- Population density affects contact intensity.
- Vaccination rate reduces disease transmission.
- Human mobility increases social interaction.
- Mask usage decreases effective contact.
- Hospital capacity characterizes healthcare availability.
- Mean age represents demographic heterogeneity.

Collectively, these variables act as observed confounders because they influence both treatment assignment and epidemic outcomes.

The use of multiple confounders ensures that causal estimation methods must appropriately adjust for covariate imbalance before estimating intervention effects.

8. Contact Intensity Model

Disease transmission is fundamentally driven by human-to-human interactions. Rather than directly assigning a transmission coefficient, the proposed simulator first estimates an intermediate behavioural variable termed the *contact intensity*. This quantity represents the average frequency of effective interpersonal interactions within a population.

The contact intensity is influenced by multiple demographic and behavioural characteristics including population density, mobility, vaccination coverage, and mask usage. These variables jointly determine the opportunities for disease transmission.

Mathematically, the latent contact score is represented as

$$x = -0.2 + 0.8D + 0.7M - 0.5K - 0.5V + \varepsilon,$$

where

- D denotes scaled population density,
- M denotes human mobility,
- K denotes mask usage,
- V denotes vaccination rate,
- ε represents random behavioural variation.

To ensure realistic values, the score is transformed using the logistic function

$$Contact = 0.2 + \frac{1}{1 + e^{-x}}.$$

The additional constant of 0.2 guarantees that even highly isolated regions maintain a small amount of unavoidable social interaction. Consequently, contact intensity lies approximately between 0.2 and 1.2.

This variable subsequently becomes the principal determinant of disease transmissibility.

9. Transmission Rate Generation

The transmission coefficient (β) represents the probability that an effective contact results in disease transmission. Rather than sampling β independently, the simulator derives it from contact intensity, thereby preserving epidemiological realism.

The latent transmission score is

$$z = -0.8 + 2.0 \times Contact.$$

The transmission coefficient is then generated using

$$\beta = 0.20 + \frac{0.65}{1 + e^{-z}} + \eta,$$

where

$$\eta \sim N(0, 0.015^2)$$

introduces stochastic environmental variation.

Finally,

$$0.15 \leq \beta \leq 0.90$$

is enforced through clipping to avoid unrealistic epidemic behaviour.

This formulation ensures that regions exhibiting larger contact intensity naturally possess larger transmission coefficients while maintaining smooth nonlinear relationships.

10. Treatment Assignment Mechanism

Unlike randomized controlled trials, interventions in observational studies are rarely assigned randomly. Government policy decisions are generally influenced by observed characteristics of the population. To reproduce this observational setting, treatment assignment is generated through a propensity score model.

Each region first receives a latent treatment score

$$S = -2.0 + 0.00025P + 1.2M - 1.5V - 0.8K + 0.8\beta + 0.6C,$$

where

- P denotes population density,
- M denotes mobility,
- V denotes vaccination rate,
- K denotes mask usage,
- β denotes the transmission coefficient,
- C denotes contact intensity.

The propensity score is obtained through logistic regression,

$$Pr(T = 1) = \frac{1}{1 + e^{-S}}.$$

Treatment is then sampled as

$$T \sim Bernoulli(Pr(T = 1)).$$

This procedure intentionally creates confounding because the same variables influencing treatment assignment also influence epidemic severity.

Consequently, estimating intervention effects requires appropriate adjustment for observed covariates, closely resembling real observational studies.

11. Compliance Mechanism

Receiving treatment does not necessarily imply that the intervention is fully implemented. In real-world public health settings, compliance varies substantially across regions because of socioeconomic conditions, behavioural adherence, and policy enforcement.

To capture this phenomenon, the simulator generates a continuous compliance score for every treated observation.

Compliance is modelled as

$$Compliance = 0.25 + 0.35K + 0.20V - 0.10M + 0.05H + \xi,$$

where

- K denotes mask usage,
- V denotes vaccination rate,
- M denotes mobility,
- H denotes hospital capacity,
- ξ represents random behavioural variability.

Compliance values are restricted to

$$0.15 \leq Compliance \leq 0.95.$$

For untreated observations, compliance is recorded as zero because no intervention exists to follow.

Introducing compliance creates treatment heterogeneity. Two regions receiving identical interventions may experience substantially different outcomes because of differences in behavioural adherence. This increases the realism of the benchmark dataset and allows evaluation of causal methods under heterogeneous treatment effects.

12. Time-Varying SEIR Epidemic Model

The spread of infection is simulated using an extended Susceptible–Exposed–Infectious–Recovered (SEIR) compartmental model. Unlike a static transmission model, the proposed framework allows the transmission coefficient to vary over time according to public health interventions and behavioural compliance.

The population is divided into four mutually exclusive compartments:

- Susceptible (S): Individuals who are at risk of infection.

- Exposed (E): Individuals who have been infected but are not yet infectious.
- Infectious (I): Individuals capable of transmitting the disease.
- Recovered (R): Individuals who have recovered and are assumed immune.

The epidemic evolves according to the following system of ordinary differential equations.

$$\frac{dS}{dt} = -\beta(t)SI$$

$$\frac{dE}{dt} = \beta(t)SI - \sigma E$$

$$\frac{dI}{dt} = \sigma E - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

where

- $\beta(t)$ denotes the time-varying transmission coefficient,
- σ denotes the incubation rate,
- γ denotes the recovery rate.

Unlike classical SEIR models where β remains constant throughout the epidemic, the proposed simulator dynamically modifies β according to intervention timing and compliance.

13. Time-Varying Transmission Dynamics

Government interventions rarely produce an immediate or permanent reduction in transmission. Instead, policy effects evolve over time due to implementation delays, behavioural adaptation, and reopening phases.

Accordingly, the simulator divides the epidemic into three phases.

Phase I: Pre-Intervention

During the first forty simulation days, no intervention is active.

$$\beta(t) = \beta_{base}$$

Phase II: Intervention Period

Between Days 40 and 120, lockdown policies reduce disease transmission.

The effective transmission coefficient becomes

$$\beta(t) = \beta_{base} \left(1 - 1.2 \times Treatment \times Compliance\right)$$

Higher compliance therefore produces stronger reductions in transmission.

Phase III: Partial Reopening

Following Day 120, restrictions are gradually relaxed.

The transmission coefficient becomes

$$\beta(t) = \beta_{base} \left(1 - 0.6 \times Treatment \times Compliance\right)$$

Finally, stochastic environmental variability is introduced through multiplicative log-normal noise,

$$\beta(t) = \beta(t) \times LogNormal(0, 0.03).$$

This stochastic component captures unpredictable day-to-day fluctuations in transmission arising from environmental conditions and behavioural randomness.

14. Potential Outcomes Framework

The benchmark dataset follows the Rubin Causal Model by explicitly generating both potential outcomes for every simulated region.

For each observation,

$$Y(0)$$

denotes the epidemic outcome in the absence of treatment,

whereas

$$Y(1)$$

represents the epidemic outcome under treatment.

Since the simulator has complete knowledge of the data generating process, both potential outcomes are observable.

Consequently, the Individual Treatment Effect (ITE) is computed as

$$ITE = Y(1) - Y(0).$$

The simulator additionally computes

$$ATE = E[Y(1) - Y(0)]$$

and

$$ATT = E[Y(1) - Y(0) \mid T = 1].$$

These quantities constitute the ground truth used for evaluating causal inference estimators.

15. Observed Outcomes

Although both potential outcomes are generated internally, only one outcome becomes observable depending upon treatment assignment.

The observed outcome is therefore defined as

$$Y = TY(1) + (1 - T)Y(0).$$

This construction faithfully reproduces the missing counterfactual problem encountered in observational studies.

Researchers analysing the dataset only observe

$$(T, Y)$$

whereas the simulator retains access to

$$Y(0),\ Y(1),\ ITE,\ ATE,\ ATT.$$

Consequently, estimation bias can be computed exactly after applying any causal inference algorithm.

16. Simulation Algorithm

Each simulated observation follows the workflow below.

1. Generate demographic covariates.

2. Compute contact intensity.

3. Generate the baseline transmission coefficient.
4. Estimate treatment propensity.
5. Assign treatment.
6. Generate compliance.
7. Initialise SEIR compartments.
8. Simulate epidemic without treatment.
9. Simulate epidemic with treatment.
10. Compute potential outcomes.
11. Compute Individual Treatment Effect.
12. Store observed outcome and all benchmark variables.

Repeating this procedure independently produces the complete benchmark dataset containing one hundred thousand simulated epidemic scenarios.

17. Benchmark Dataset Description

The final benchmark dataset consists of 100,000 independently simulated epidemic scenarios. Each observation corresponds to a hypothetical geographical region with its own demographic characteristics, epidemiological parameters, intervention assignment, and epidemic trajectory.

Unlike conventional epidemiological datasets, the proposed benchmark additionally stores the true potential outcomes and treatment effects generated internally by the simulator. Consequently, the exact causal effect is known for every observation, enabling quantitative evaluation of causal inference estimators.

The dataset contains a total of 24 variables, each serving a specific role within the simulation framework.

18. Variable Dictionary

Variable	**Description**
population_density	Population density of the simulated region. Larger values generally increase interpersonal contact.

vaccination_rate	Proportion of vaccinated individuals in the population.
mobility	Relative movement of individuals across the region.
mask	Average mask compliance before intervention.
hospital_capacity	Healthcare infrastructure represented through available hospital capacity.
age_mean	Average age of the population.
contact_intensity	Behavioural measure representing effective social interaction generated from demographic variables.
beta	Baseline disease transmission coefficient.
gamma	Recovery rate used in the SEIR model.
sigma	Incubation rate of exposed individuals.
treatment	Binary intervention indicator.
propensity_score	Probability of receiving treatment estimated through the observational assignment mechanism.
compliance	Behavioural adherence to the assigned intervention.
lockdown_strength	Intensity of intervention applied to treated regions.
R0	Basic reproduction number computed as β/γ.
Y0_peak	Peak infection obtained under the untreated scenario.
Y1_peak	Peak infection obtained under the treated scenario.
ITE_peak	Individual treatment effect computed as $$Y(1) - Y(0).$$
peak_infection	Observed peak infection according to actual treatment assignment.
peak_day	Simulation day at which peak infection occurs.
area_under_curve	Area under the infectious curve representing cumulative epidemic burden.
mean_infected	Average infectious proportion throughout the simulation.
max_exposed	Maximum exposed population during the epidemic.
final_recovered	Recovered proportion at the end of the simulation.

19. Characteristics of the Benchmark Dataset

Several properties were intentionally incorporated into the benchmark dataset in order to resemble realistic observational data.

- Treatment assignment is confounded rather than randomized.
- Compliance varies across treated observations.

- Disease transmission evolves dynamically over time.
- Intervention effects differ between observations.
- Potential outcomes are completely known.
- Behavioural variables influence both treatment assignment and epidemic outcomes.
- Stochastic variability prevents deterministic epidemic trajectories.

These characteristics create a challenging environment for evaluating modern causal inference algorithms.

20. Applications

The proposed benchmark dataset is suitable for evaluating a wide variety of causal inference methodologies including

- Propensity Score Matching
- Inverse Probability Weighting
- Stabilized IPW
- Augmented IPW
- Doubly Robust Estimation
- Regression Adjustment
- DoWhy Framework
- EconML
- Meta Learners (S, T and X Learners)
- Double Machine Learning
- Causal Forests

Because the true treatment effects are known, estimation bias and mean squared error can be computed directly.

21. Advantages of the Proposed Benchmark

Compared with existing synthetic epidemiological datasets, the proposed simulator offers several advantages.

1. Time-varying disease transmission.
2. Behaviour-driven contact intensity.
3. Observational treatment assignment.
4. Partial compliance.
5. Heterogeneous treatment effects.
6. Known counterfactual outcomes.
7. Realistic epidemic dynamics.
8. Large-scale simulation (100,000 observations).
9. Direct suitability for benchmarking causal inference algorithms.

22. Limitations

Although the simulator incorporates multiple realistic components, several simplifying assumptions remain.

- Homogeneous mixing within each simulated region.
- No explicit spatial interaction between regions.
- Fixed compartmental structure.
- No demographic births or natural deaths.
- Simplified behavioural dynamics.

These assumptions were adopted to balance epidemiological realism with computational efficiency.

23. Future Work

Several extensions may further improve the benchmark.

- Multi-wave epidemic dynamics.
- Spatially connected regions.
- Multiple competing interventions.
- Vaccination booster campaigns.
- Time-varying behavioural adaptation.
- Hidden confounders for sensitivity analysis.
- Instrumental variable generation.
- Dynamic treatment regimes.
- Reinforcement learning based intervention policies.

24. Conclusion

This report presented the complete methodology used to construct a synthetic epidemiological benchmark dataset for evaluating causal inference methods.

The simulator integrates demographic heterogeneity, behavioural characteristics, observational treatment assignment, intervention compliance, and a time-varying SEIR epidemic model into a unified simulation framework. Unlike conventional observational datasets, the proposed benchmark provides complete knowledge of the underlying causal structure, enabling rigorous quantitative comparison of causal inference algorithms.

The resulting benchmark dataset is computationally scalable, statistically realistic, and specifically designed for methodological research in modern causal inference.

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